Ontologies, Multiagent Digital Twin and Intelligent Telemetry for Internet Infrastructure (DT-II)

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Knowledge Base (Ontology)

The Knowledge Base

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Knowledge Base (Ontology)

Description Logic (OWL)

The Knowledge Base

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The Knowledge Base

Knowledge Base (Ontology)

Description Logic (OWL)

TBox
Terminological Box

ABox
Assertion Box

RBox
Role Box

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Knowledge Base (Ontology)

The Knowledge Base

Description Logic (OWL)

TBox

Terminological Box

Concept Definitions

- Person ⊑ Mammal[(1)]
- Car ⊑ Vehicle[(2)]
- Driver ≡ Person who drives a vehicle

ABox

Assertion Box

Facts about individuals

- Mary is a Person
- Fiat is a Car
- Mary drivex Fiat

RBox

Role Box

Relationship Rules

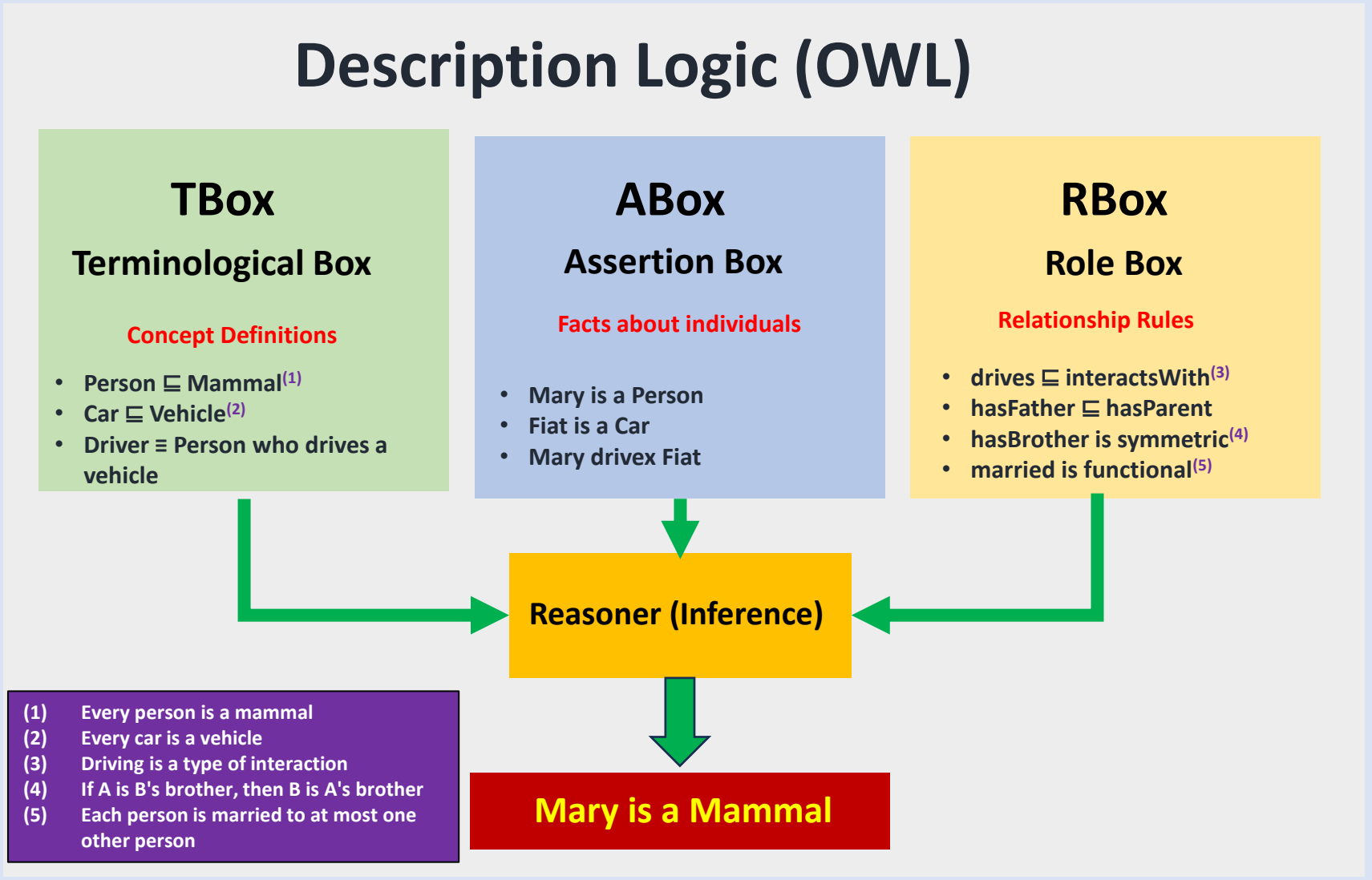
- drives ⊑ interactsWith[(3)]
- hasFather ⊑ hasParent
- hasBrother is symmetric[(4)]
- married is functional[(5)]

(1) Every person is a mammal
(2) Every car is a vehicle
(3) Driving is a type of interaction
(4) If A is B's brother, then B is A's brother
(5) Each person is married to at most one other person

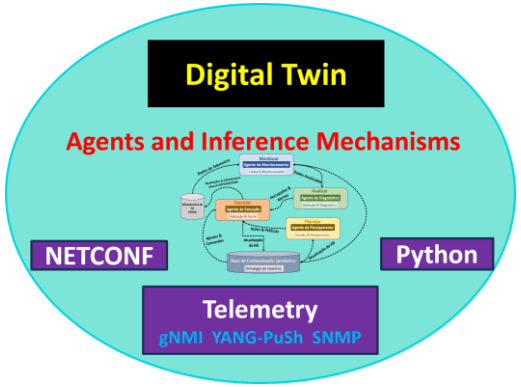
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Knowledge Base (Ontology)

The Knowledge Base



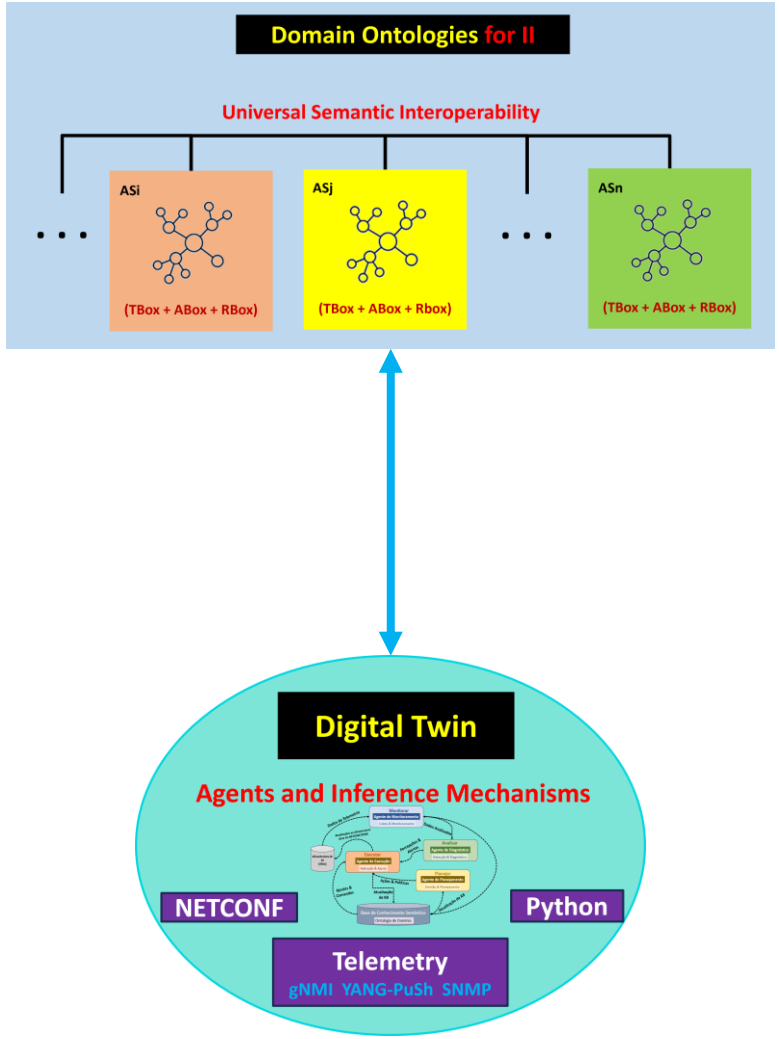
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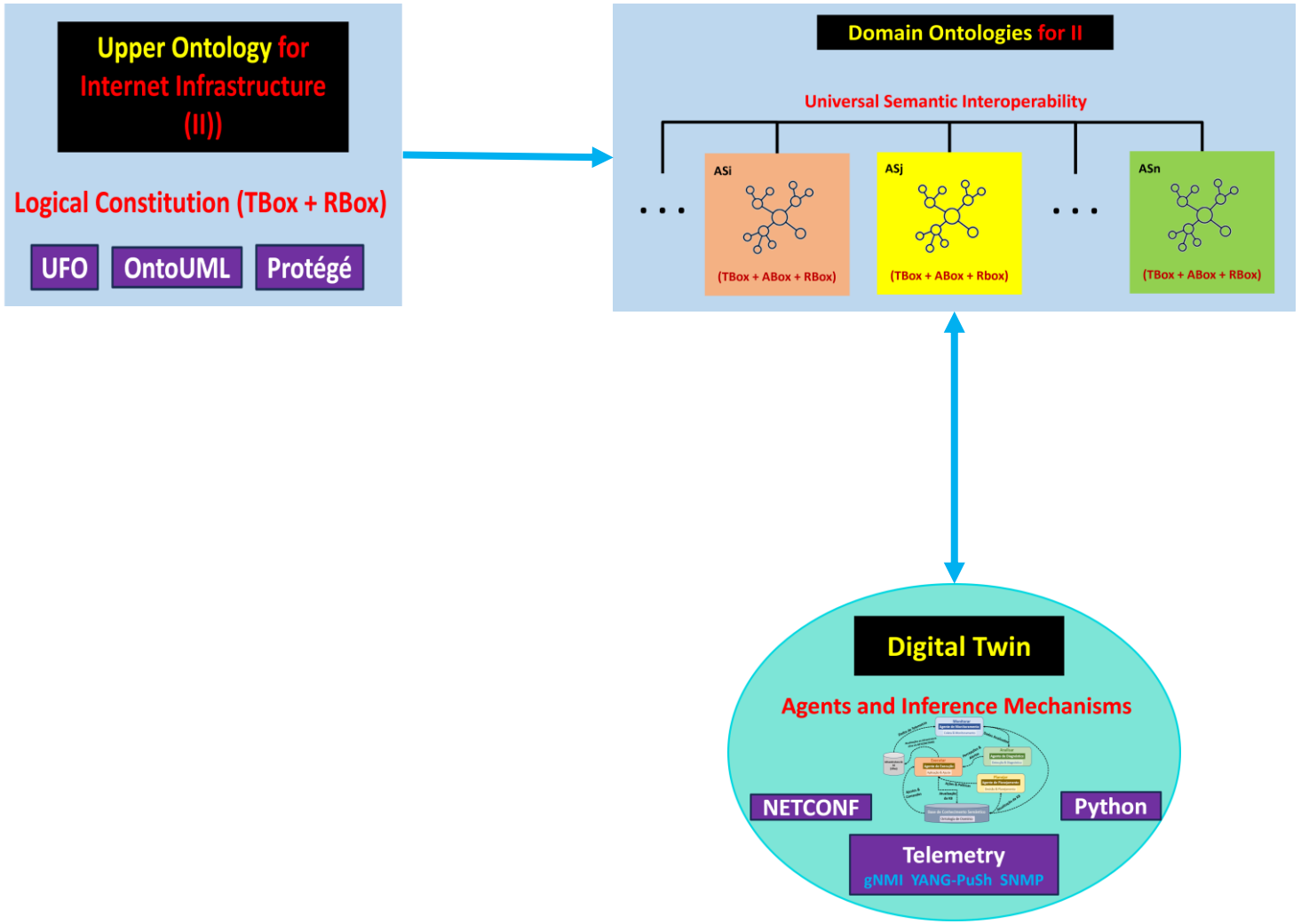
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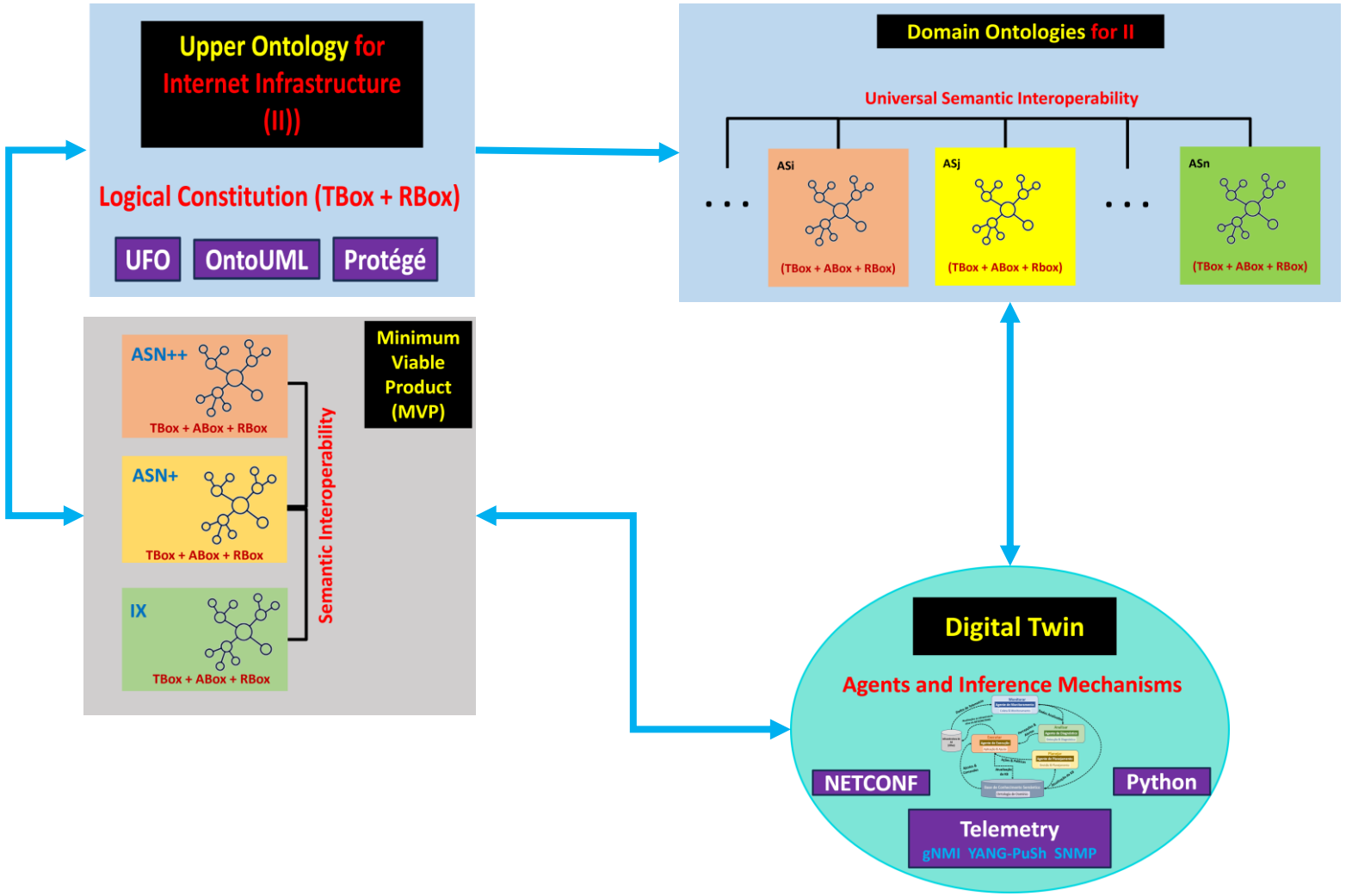
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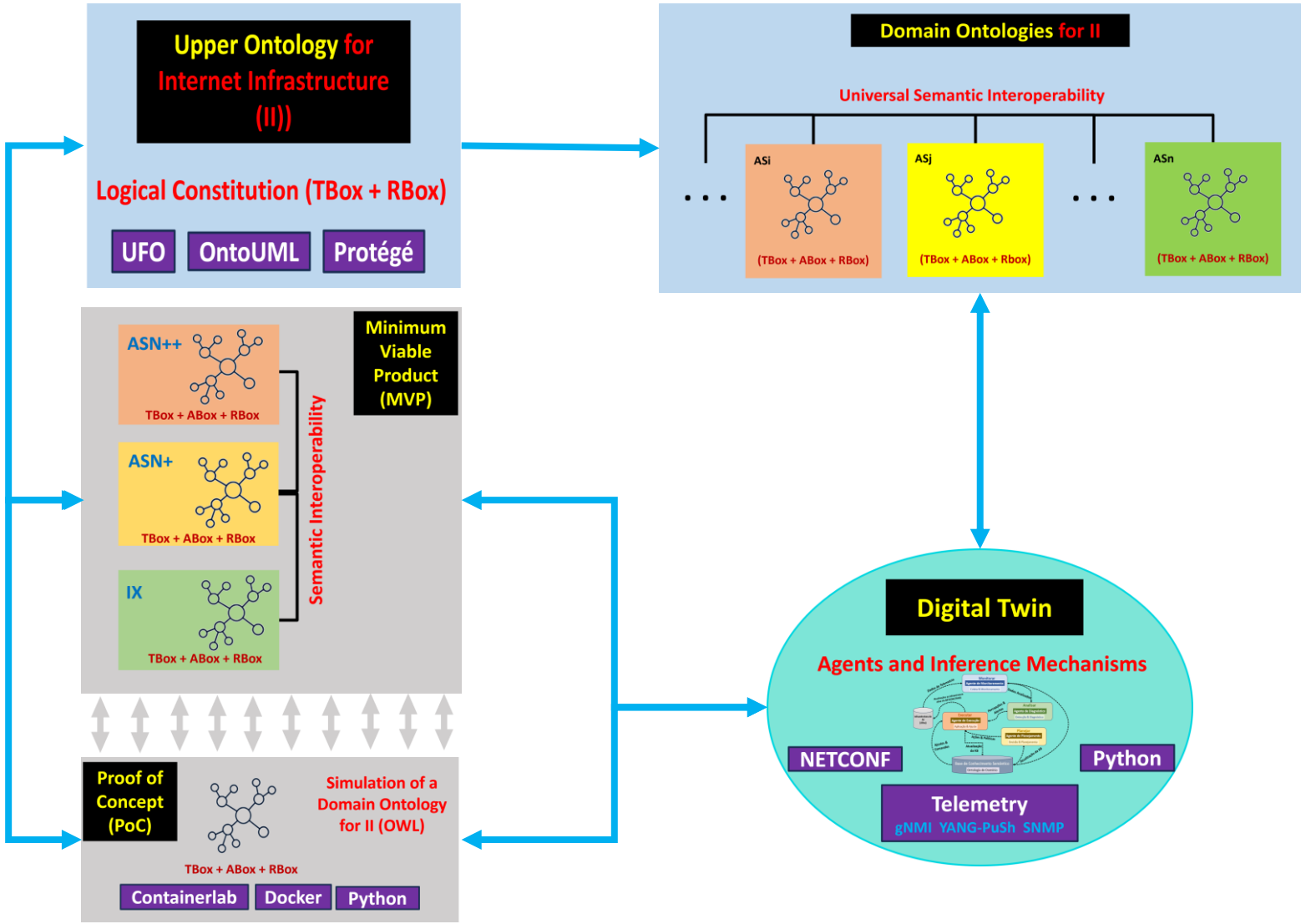
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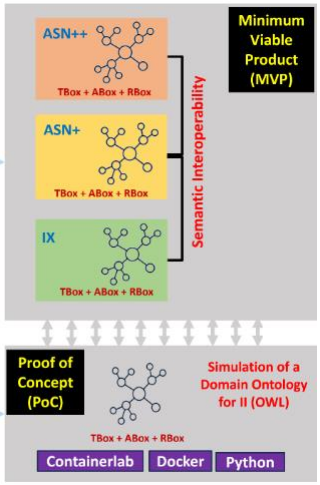
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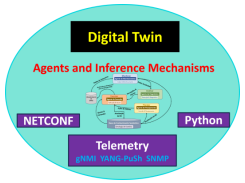
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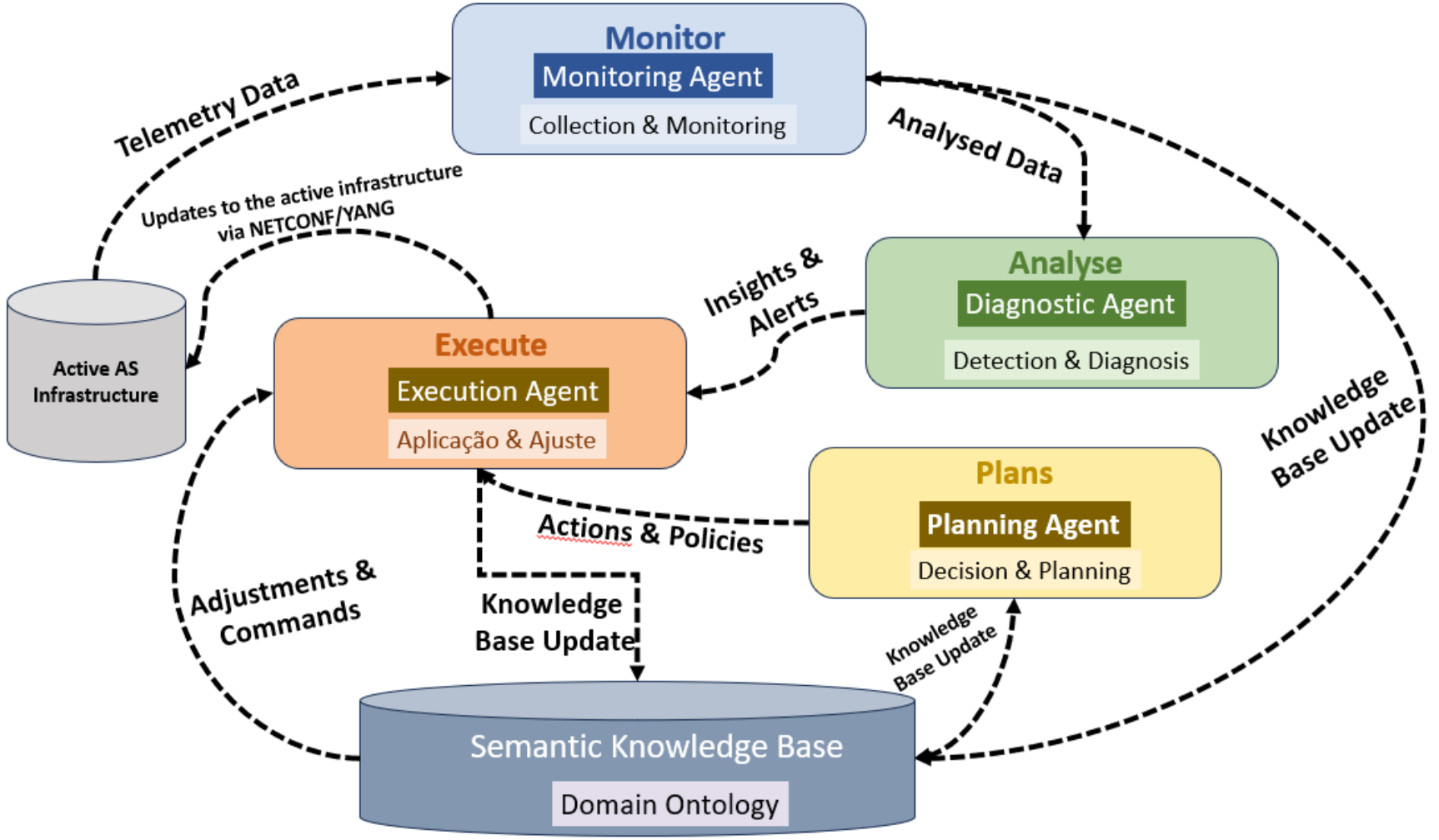
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Classification of MVP participants, by level of complexity

High Complexity (ASN++)	Medium Complexity (ASN+)	Special Domain (IX)
Meet highly specialised and critical demands	Meet intermediate demands, with some level of specialisation	Meet a highly complex environment that centralises the interests of a large number of ASes of varying complexity
Robust infrastructure and advanced technology	Adequate infrastructure, but less sophisticated	Very complex infrastructure with highly diverse Internet traffic
Highly qualified multidisciplinary teams	Qualified technical teams, but less diverse	Highly qualified and diverse multidisciplinary teams
Services with high availability, redundancy, and security	Reliable services, but with less redundancy	Services with high availability and differentiated within the same environment
Integration with complex, large-scale systems	Moderate customisation capacity	Little or no integration capacity
Serve large corporations, public bodies, or strategic sectors	Serve medium-sized companies	Serve small, medium, large, and very large institutions with diverse service/-traffic demands
Bring the complexity of global routes, transit policies, and complex peering agreements	Bring the edge perspective, where regulations such as LGPD/SLA and local access policies apply	The neutral traffic exchange point, where the semantics of relators (multilateral BGP sessions, exchange agreements) are tested to the limit





MAPE-K

Jeffrey O. Kephart and David M. Chess. “The Vision of Autonomic Computing”. In: Computer. Vol. 36. 1. IEEE,2003, pp. 41–50. doi: 10.1109/MCOM.2003.1160055

DT Reference Architecture

Ontological Layer	**Tbox:** define core classes (**NetworkNode, Connection, Channel**) and constraints
Identity Layer	Assigns stable IRIs to instances, separating identity from state
Ingestion Layer	Collects telemetry (**YANG-Push, gNMI, SNMP, syslog**) and normalises data into the **Node–Connection–Channel** triad
Persistence Layer	**ABox + Rbox:** stores facts in RDF/OWL and maintains temporal history via observations
Validation Layer	Applies structural constraints and logical consistency checks with **DL and DeL** reasoners
Reasoning Layer	Infers states and propagates failures
Exposure Layer	Provides SPARQL queries, dashboards, alerts, and integration with NOC/orchestration systems

Use Cases (I)

Ontological Compliance Regime and Logical Governance in DT-II

The Ontological Compliance Regime is the governance mechanism of **DT-II** that establishes authority and policy compliance over digital entities based on meaning, context, and relations — transcending traditional geophysical boundaries. In this model, jurisdiction ceases to be a static hardware attribute and becomes a **normative entity mediated by relators**.

Foundation and Axiomatisation

Defined in the ***TBox*** through restriction axioms, the Ontological Compliance Regime employs the Reasoner to determine which norms (e.g., LGPD, GDPR, SLAs) apply to real-time data flows. In the ***RBox***, property chains formalise propagation: if a `NetworkNode` is linked to a legal norm, all `Channels` and `Connections` mediated by relators inherit the same constraints.

Role of the Digital Twin

The Digital Twin acts as the guardian of jurisdiction. By ingesting telemetry (gNMI, YANG-Push, SNMP), the system populates the ***ABox***. Through **Semantic Elevation**, it verifies, for example, whether a health data packet traverses a node under a jurisdiction that does not meet the privacy requirements defined in the ***TBox***.

Operationalisation and Resilience

In dynamic events (e.g., DDoS attacks, BGP failures), the Ontological Compliance Regime guides the **MAPE-K** cycle: (a) **Analysis**: the reasoner detects violations of jurisdiction axioms; (b) **Planning**: the system seeks compliant alternatives; (c) **Execution**: configuration is applied via ***NETCONF*** only after logical validation.

In summary, the Ontological Compliance Regime transforms infrastructure into a juridically aware system, wherecompliance is mathematically verified by DL, ensuring governance integrity in complex and mutable networks.

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Use Cases (II)

Model Behaviour under a DDoS Attack

In a Distributed Denial of Service (DDoS) attack, response time is critical. The Digital Twin acts as a Semantic Defence System, leveraging the ***TBox***, ***RBox***, and ***ABox*** within the ***MAPE-K*** cycle as follows:

- **Monitoring** Telemetry (gNMI, YANG-Push, SNMP) updates the ***ABox*** in real time. For example, a traffic spike from 1 Gbps to 100 Gbps is recorded as:

```
inst:Unusual_Flow a ii:AnomalousFlow
```

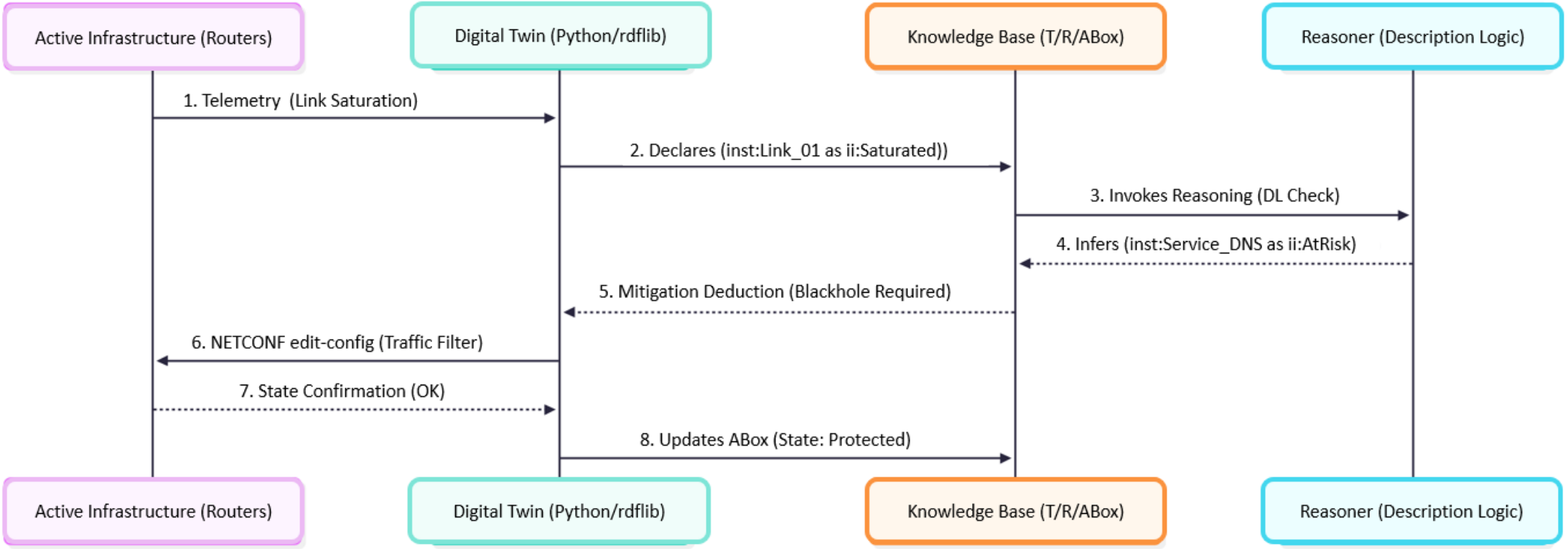
- **Analysis** The Reasoner applies property chains from the ***RBox***: (a) the anomalous flow saturates `Channel_Alpha`; (b) transitivity propagates risk to dependent services (e.g., DNS); (c) the origin is traced to a specific **Ontological Compliance Regime**.
- **Planning** Consulting the ***TBox***, the system applies rules: if a critical service is at risk and the origin is external, mitigation (Blackholing or Rate Limiting) must occur at the edge. The plan prioritises nodes sustaining the network's **Ontological Normative Base**.
- **Execution** Logical decisions are translated into technical commands. Via ***NETCONF***, edge routers are reconfigured to filter malicious traffic.
- **Advantages of the Semantic Approach** Unlike conventional defences that block IPs, **DT-II** enables: (a) **Context Awareness**: cooperation agreements mapped in the ontology may trigger governance APIs instead of simple blocking; (b) **Mereological Resilience**: the Reasoner may sacrifice non-vital channels to preserve critical services; (c) **Traceability**: each mitigation step is explainable, linked to axioms in the ***TBox***; (d) **Compliance**: no action is executed without validation by the ***Ontological Normative Base***.
- **Sequence Flow** Mitigation occurs within milliseconds, moving from the **Active Infrastructure** to the **Logical Core** and back. Figure 4 illustrates the process.

In summary, the ontology-based approach enables policy-driven mitigation, where assets of higher semantic andcontractual value are prioritised, ensuring resilience and governance integrity.

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Use Cases (II)



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Contributions

- an ontology-driven reference model for II centred on Node–Connection–Channel, grounded in UFO anddeveloped in OntoUML;
- an operationalisation in DL/OWL supporting automated reasoning over TBox,ABox, and RBox;
- a synchronisation approach aligning the KG with model-driven telemetry (e.g., YANG-Push) for near real-timestate updates;
- a governance mechanism based on an Ontological Compliance Regime and SLA constraints, enablingauditable and explainable compliance checks; and
- a multi-agent Digital Twin architecture following the MAPE-K cycle (Kephart and Chess 2003; IBMCorporation 2005) for policy-driven actions.

Project Documentation

Project Digital Twin for Internet Infrastructure

Main: https://github.com/LJuliaoBraga/dtii

English: https://github.com/LJuliaoBraga/dtii/blob/main/docs/en-us/2026-ProjectDT-II-EN.pdf
Portuguese: https://github.com/LJuliaoBraga/dtii/blob/main/docs/pt-br/2026-ProjectDT-II-PT.pdf

ANMS 2006: https://github.com/LJuliaoBraga/dtii/blob/main/docs/events/anms%202026/2027_anms.pdf
ANMS 2006 Presentation: https://github.com/LJuliaoBraga/dtii/blob/main/docs/events/anms%202026/ANMS Presentation.pdf

Thank you

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